

# EMERGENCE LAB | EXPERIMENT 07

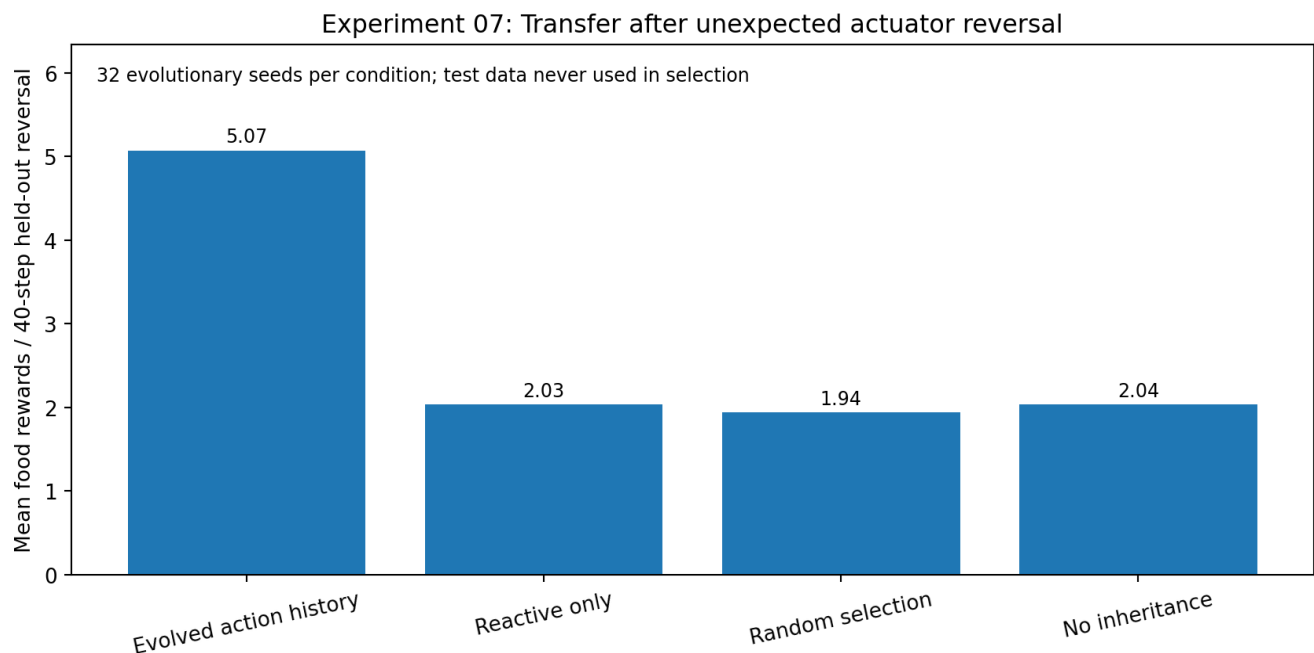
## Evolving action-conditioned adaptation under unknown motor effects

Exploratory computational research note · 2026-10-08 · Not peer-reviewed or publicly preregistered

**Question.** Can evolution favor information-processing controllers that adapt to unexpected changes in the effects of their own actions?

**Design.** 32 evolutionary seeds per arm; 64 candidate controllers; 90 generations; each controller chooses its own left/right actions and encounters noisy food signals and unknown actuator polarity. Final selection uses separate development lifetimes; evaluation uses unseen episodes where actuator direction reverses halfway through.

**Important boundary.** The memory unit and its action-conditioned input were engineered by us. We measure adaptive behavior, not consciousness.



**Primary comparison.** Evolved memory: 5.072 vs reactive: 2.034 food rewards per 40-step reversal episode. Mean seed-paired difference: 3.038, 95% percentile bootstrap CI [2.200, 3.891]. Criterion met.

# METHODS AND RESULTS

| Condition              | Unseen reversal | Stable world | Reversal survival |
|------------------------|-----------------|--------------|-------------------|
| Evolved action history | 5.072           | 6.475        | 0.899             |
| Reactive only          | 2.034           | 4.207        | 0.751             |
| Random selection       | 1.938           | 3.175        | 0.757             |
| No inheritance         | 2.038           | 3.499        | 0.768             |

**Full minus reactive:** 3.038 [95% CI 2.200, 3.891], with 25/32 seeds favoring full.

**Full minus random selection:** 3.134 [95% CI 2.243, 3.993], with 28/32 seeds favoring full.

**Full minus no inheritance:** 3.034 [95% CI 2.152, 3.880], with 30/32 seeds favoring full.

**Full minus scrambled:** 3.741 [95% CI 2.743, 4.722], with 30/32 seeds favoring full.



Training data, test data, and random seeds are included in the accompanying ZIP. Code and protocol SHA-256 hashes were recorded before the primary run.

# INTERPRETATION & LIMITATIONS

## What this shows

Selection in a designed artificial world can favor a controller that uses action-conditioned feedback. The intervention is informative about task performance, not the presence of subjective experience.

## What we built in

We specified the reward rules, sensorimotor feature, memory update architecture, genetic mutation process, and surprise reversal. Agents did not invent neurons, bodies, or the concept of self.

## What remains unresolved

This is a one-dimensional synthetic benchmark with a limited controller family. Reactive and memory arms differ in representational capacity. Test-time scrambling also alters downstream pseudorandom-number draws, so this ablation is not perfectly paired.

## Publication status

This is suitable for an openly labeled exploratory computational report, not a peer-reviewed finding about consciousness. Independent replication, broader environment families, and a PUBLIC pre-run registration would be necessary for stronger claims.

## Sources

Bongard, Zykov & Lipson (2006), *Science*, doi:10.1126/science.1133687; Nguyen et al. (2020), arXiv:2011.12860; Kahl et al. (2021), arXiv:2112.05577; Center for Open Science simulation preregistration guidance.

Replication: `python -m unittest -q test_engine07.py; python engine07.py --seeds 32 --seed-start 870000 --out primary`